

Assessment Task 1: Practical & Written Assignment (20%).

Course code and name: COSC2753 Machine Learning

Length: No more than 05 pages (excluding Cover Page, References List) plus up to 2 pages for appendices, font size 12

Type: Individual

Feedback mode: Written feedback

Late work: For assignments 01 to 05 days late, a penalty of 20% (of the marks awarded) per day will apply. For assignments more than 05 days late, a penalty of 100% will apply. Weekend days (Saturday and Sunday) are counted when counting total late days.

Learning Objectives Assessed

- **CLO1:** Understand the fundamental concepts and algorithms of machine learning and applications
- **CLO3:** Set up a machine learning configuration, including processing data and performing feature engineering, for a range of applications
- **CLO4:** Apply machine learning software and toolkits for diverse applications

Ready for Life and Work

Completing this assessment would help you achieve the following employability skills:

- Analyse and model requirements and constraints for the purpose of designing and implementing solutions to a learning challenge
 - Evaluate and compare approaches and algorithms on the basis of the nature of the problem/task being addressed
 - Interpret abstract theoretical propositions, choose methodologies, justify conclusions and defend professional decisions to both IT and non-IT personnel via technical reports of a professional standard and technical presentations
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Assessment Details

In this assignment you will explore a real dataset to practice the typical machine learning process. This assignment is designed to help you become more confident in applying machine learning approaches to solving tasks.

- Assignment 01 specifications is here [Machine Learning COSC2753_2026B_Assignment 1.pdf](#)

---> update the new version on 14-July-2026 [Machine Learning COSC2753_2026B_Assignment 1_update_version.pdf](#)

- The dataset for A1 is here:

- [2026B_dataset.zip](#) ↓
- [meta_data.pdf](#)

Support Resources

This assessment requires that you meet RMIT's expectations for academic integrity. More information and advice on how to avoid plagiarism are available in the Getting Started module.

Open [the academic integrity page](#).

Additional library and learning resources are available to help with the assessment in this course

Link to [Assignment Support](#).

Submission Instruction

You will need to submit the 04 following deliverables:

Please note that you must submit all the 04 deliverables as required. However, your final grade will be based solely on your PDF report, your prediction file and presentation. That means any part of your code that is not covered or explained in your report and presentation will not be considered for grading.

- **A PDF report** *preferably converted form of notebook* used for the model development including critical analysis of your approach and ultimate judgment. The report should be in PDF format. Search for instructions on converting the notebook to PDF. Please name the report by following this convention: *COSC2753_A1_YourStudentID*
 - Please submit **ONLY your PDF REPORT** file to this page.
- **A Video presentation** around *05-7.5 minutes (minimum of 05, and maximum of 7.5 minutes)*. You should not exceed 7.5 minutes, as you will be marked only based on the first 7.5 minutes of your presentation. You will lose mark if it is less than **05 minutes**.
- **A Set of predictions** from your ultimate judgment. Should be in CSV format. Note that "s1234567_predictions.csv": shows the expected format for your predictions on the unseen test data, please **do NOT change format or order of this file**.
 - Name your prediction file as *COSC2753_A1_Predictions_YourStudentID.csv*
 - Please submit your prediction to the separate Canvas page - Prediction submission.
- **Your Python scripts or Jupyter notebooks** used to perform your analysis.
 - Zip your Jupyter notebook file and other necessary files before submitting it
 - Should be a ZIP file containing source code and all the support files. Make sure that *your assignment can run only with the code included in your zip file!*
 - Name your zip file as convention below.
 - Please make sure that the notebook must be readily opened and executed on any standard machine/setting. We strongly recommend you to attach a README file with instructions on how to set up the environment to run your file (e.g., put data files in which folder, install which libraries/packages etc.) and how to run your application.
 - Please note that your code will be checked for plagiarism by our specialised software and both by Turnitin.
 - Please submit your code (zip) file to the separate Canvas page - Code submission.

Please name your report, video and source code by following this convention:

COSC2753_A1_YourStudentID

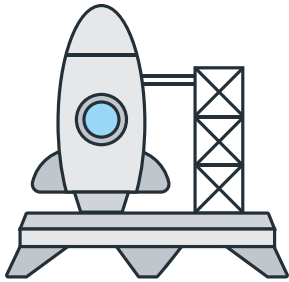
Rubric

You can review the rubric for this assignment before making a submission to Turnitin.

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Criteria	Ratings						Pts
Approach 1) Exploratory Data Analysis (EDA) leading to well informed approach. 2) Identifying an adequate evaluation framework that is tailored to the problem. 3) Well justified method/algorithms selection. 4) Hyper parameters selection strategy. 5) Through analysis of the models. 6) Approach satisfy all the requirements and restrictions.	35 to >31.5 pts HD+ Outstanding across the course.	31.5 to >28.0 pts HD The approach is an excellent and extremely thorough investigation of the chosen ML problem. All elements adequately analysed. Goes beyond what is done in class.	28 to >24.5 pts DI The approach is a good and reasonably thorough investigation of the chosen ML problem. There are small gaps between in the investigation in what could have been explored. Goes beyond what is done in class.	24.5 to >21.0 pts CR The approach is sufficient, but not a thorough investigation of the chosen ML problem. There are gaps in the investigation and alternative algorithms or techniques are better than the ones in the approach. The approach has a limited consideration of the unique aspects of the chosen ML problem.	21 to >15.0 pts PA The approach is a minimally sufficient investigation of the chosen ML problem. It only examines the bare minimum requirements of suitable techniques and algorithms. There are many gaps in the investigation and there are algorithms or techniques are clearly more suited to the chosen ML problem.	15 to >0 pts NN Poor, superficial, or incomplete approach that does not meet the minimum requirements for PA.	35 pts
Ultimate Judgement & Analysis 1) Analysis of the model and the outputs using suitable methods. 2) Make a clear ultimate Judgment. 3) Rational behind the ultimate model is clear and considers all the aspects. 4) Limitations of the model identified.	20 to >18.0 pts HD+ Outstanding across the course.	18 to >16.0 pts HD Ultimate Judgement is established and exceptionally justified. Evaluation of the Ultimate Judgement is exceptional and clearly demonstrated the viability of the trained model in real-world practice and limitations.	16 to >14.0 pts DI Ultimate Judgement is established and suitably justified. Evaluation of the Ultimate Judgement is sound and suitably explained, however, the reader may not be fully convinced and have minor questions.	14 to >12.0 pts CR Ultimate Judgement is established, but there are unexplained choices, or the justification is hard to follow. An sufficient attempt at evaluating the Ultimate Judgement is made.	12 to >9.0 pts PA An Ultimate Judgement is made by not justified.	9 to >0 pts NN An Ultimate Judgement is not made.	20 pts

Criteria	Ratings						Pts
Test Performance Performance on the unseen test set	15 to >13.5 pts HD+ Outstanding across the course (Top 5%).	13.5 to >12.0 pts HD In Top 5-20%.	12 to >10.5 pts DI In Top 20-25%.	10.5 to >8.5 pts CR In Top 25-50%.	8.5 to >6.5 pts PA Better than random model but not in top 50%.	6.5 to >0 pts NN Less accurate than a random model.	15 pts
Implementation 1) Code is well documented and easy to understand. 2) Code does not contain errors. 3) Code contain evidence of all investigations mentioned in report. 4) Code is optimal and shows good programming practices.	15 to >13.5 pts HD+ Outstanding across the course.	13.5 to >12.0 pts HD Code is exceptional and satisfy all the elements.	12 to >10.5 pts DI Code is styled and organised reasonably. Commenting could be improved.	10.5 to >8.5 pts CR Code is styled and organised reasonably. Commenting could be improved. Few minor errors.	8.5 to >6.5 pts PA Code is styled and organised poorly, not following general good programming practices. Commenting is rare. Implementation has minor issues but works.	6.5 to >0 pts NN Code is styled and organised poorly, not following general good programming practices. Contain major errors.	15 pts
Report and Presentation 1) Report is easy to read and well structured. 2) Presentation is well structured, easy to follow and exclusive.	15 to >13.5 pts HD+ - The report is exceptionally well-structured, clear, and easy to read, with excellent flow and logical organization. - Presentation is highly engaging, professionally delivered, easy to follow, and visually appealing. - Exclusive insights and creativity enhance the overall impact.	13.5 to >12.0 pts HD - The report is well-structured, clear, and easy to read, with only minor readability issues. - Presentation is well-organized, engaging, and easy to follow. - Strong delivery with effective supporting materials.	12 to >10.5 pts DI - The report is mostly clear and structured, though some sections may lack coherence or detailed explanation. - Presentation is structured and understandable but could be more engaging. - Some aspects of delivery, clarity, or visuals need improvement.	10.5 to >8.5 pts CR - The report has structure but may contain readability issues, inconsistencies, or lack of depth. - Presentation lacks clarity in some parts, making it harder to follow. - Engagement and delivery need improvement, with some disorganized elements.	8.5 to >6.5 pts PA - The report is minimally structured but lacks clarity and logical flow. - Presentation is difficult to follow, lacks engagement, and has weak organization. - Basic effort is shown, but significant improvements are needed in content and delivery.	6.5 to >0 pts NN - The report is poorly structured, difficult to read, and lacks logical organization. - Presentation is unclear, unstructured, and ineffective in conveying key points. - Minimal or no effort in ensuring readability, engagement, and professionalism.	15 pts
Total Points: 100							